University of Florida, Department of Mathematics

Algebra First-Year Exam, August 2019, Part 1

Answer four problems. (If you turn in more, the first four will be graded.) Within reason, you may quote theorems as long as you state them clearly.

1. Let p be a prime, let G be a finite group, and let P be a Sylow p-subgroup of G. Let H be a subgroup of G containing the normalizer $N_G(P)$ of P in G. Prove that then $N_G(H) = H$.

2. State and prove the Second Isomorphism Theorem for groups.

3. Let $n \geq 2$, and let S_n be the symmetric group on n letters.

(a) Define what is a transposition of S_n.

(b) Give an example of a transposition of S_n.

(c) Prove that every element of S_n is a product of transpositions of S_n.

4. Let G be a finite group, and assume that G acts transitively on the finite non-empty set Ω. Let $\omega \in \Omega$. We denote by G_ω the set of elements of G that fix ω.

(a) Prove that G_ω is a subgroup of G.

(b) Prove that the number of elements in Ω is exactly $[G : G_\omega]$.

5. Let G be a finite group.

(a) Define what it means to say that G is solvable.

(b) Without assuming any result about solvable groups (i.e. just from your definition), prove that if the order of G is 20 then G is solvable.

6. Prove or disprove the following statement. Let n be a natural number, let S_n be the symmetric group on n letters, let A_n be the alternating group on n letters, and let $\sigma \in S_n$. Then, if σ has order 2019 then $\sigma \in A_n$.